

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed University)

M.Tech/MS/PhD Examinations

Assessment – I - (Spring 2022)

Course Name: Data Science Tools & Techniques

Course Code: AM624D

Date: 07 – 03 - 2022

Duration: 01 Hr

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly. All diagrams (if any) to be neatly labeled. Attempt all questions. Calculators are allowed.
2. Before attempting please ensure you have the correct question paper.

Q1	WAP in python to multiply two square matrices using for loops without using inbuilt functions for matrix multiplication.	[2.5]
Q2	Write a program to input number of lines from the user and print the pattern: If the user enters 4 following patters should be printed: * ** *** ****	[2.5]
Q3	For the following 3D array: a3 = np.array([[[10, 11, 12], [13, 14, 15], [16, 17, 18]], [[20, 21, 22], [23, 24, 25], [26, 27, 28]], [[30, 31, 32], [33, 34, 35], [36, 37, 38]]]) - a) extract [23, 24, 25] using slicing b) extract the highlighted value using negative indexing	[2.5]
Q4	Write a single line script using map & lambda to print a list containing the power of the said numbers: bases raised to the corresponding number in the idx lists without using pow() function. bases = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100] idx = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10] Expected output: [10, 400, 27000, 2560000, 312500000, 46656000000, 82354300000000, 16777216000000000, 387420489000000000, 100000000000000000000]	[2.5]

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ALL THE BEST

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